

SOC'26 - American Put Option Pricing

Week 1 Assignment

By Aayush Gupta (25B0064)

Investment Context :-

The client owns 500 shares with current spot price of ₹2,450 of a large cap stock (stock of companies with more than 10 billion USD of market cap) and is expecting a 10-15% fall in the value of the stock over the period of upcoming three months. The client does not wish to sell his shares due to tax and conviction reasons. Thus the objective is to protect the downside risk while maintaining ownership of shares.

Call vs Put :-

As this is a downside hedge, I would advise the client to purchase put options on the same large-cap stock. This is because put options are fundamentally bearish in nature, meaning they increase in value when the stock price declines. The reason for this is that a put option gives the holder the right to sell shares at a predetermined price (strike price), thereby helping offset losses in the underlying stock during a market correction.

The other type of option contract, namely call options, provides the right to buy shares at a fixed price and generally benefits when stock prices rise. Therefore, call options are not suitable for protecting against downside risk in this situation.

In the event that the stock price increases instead of falling, the primary downside of this strategy would be the loss of the premium

paid for the put option contract. However, this loss would likely be partially or fully offset by gains in the stock position, allowing the client to retain upside participation while still having downside protection in place.

Example :-

Current stock price = **₹2450**

Strike price(K) = **₹2450**

Put/Call premium = **₹100/share**

Contract maturity = **1 year**

Client horizon = **3 months**

Stock falls by **15% after 3 months**

Then, Stock price(S) = **₹2,082.5**

Exercising the put contract will give $\max(K-S, 0)$

i.e. $₹367.5 - 100 = \textbf{₹267.5}$ per share or more (depending on value of put contract)

This amount helps offset most of the losses that occur in the value of the stock.

On the other hand, buying call option contract –

Profit/loss per share = $\max(S-K, 0) = \max(-367.5, 0) = \textbf{₹0}$

So the maximum loss would be the premium paid for the contract plus the loss of the value of the stock.

Strike Price(K) for Contract :-

For this hedge, I would recommend an At The Money (ATM) put option with a strike price (K) of ₹2450, equal to the current stock price. Although an Out of money (OTM) put would be cheaper, it would only provide protection after a larger fall in stock price. Since

the client is specifically concerned about a potential 10–15% correction, an ATM put would provide stronger downside protection from the beginning of the decline.

At the current spot price of ₹2450, the put option would be classified as an ATM put because the strike price is equal to the stock price. As a result, the intrinsic value (value if contract is exercised) of the option would initially be zero [$\max(2450 - 2450, 0) = 0$], meaning most of the premium paid would consist of time value. This time value reflects the possibility that the stock price may decline before the option expires.

European vs American :-

There is one major difference between American and European option contracts which is that European contracts can only be exercised at time of expiry while American ones don't have this constraint.

For this hedge, an American put may be preferable because it provides the flexibility to exercise before expiry if the stock falls sharply and the put becomes deeply in-the-money (ITM). However, early exercise is generally not optimal because the option may still retain time value. In extreme downside scenarios, immediate exercise could become beneficial, particularly when receiving cash earlier allows reinvestment at prevailing interest rates.

Payoff table :-

[Excluding premium paid on Put contract]

Spot at Expiry (S)	Put Payoff $=\max(K-S, 0)$	Portfolio Outcome (Stock + Long Put)
₹2100	₹350	₹2450
₹2300	₹150	₹2450

₹2450	₹0	₹2450
₹2600	₹0	₹2600
₹2700	₹0	₹2700

Final Trade idea :-

I would recommend purchasing an at-the-money (ATM) put option on the same large-cap stock with a strike price of ₹2450 and a maturity of one year. A stock put would be preferable to a Nifty or Bank Nifty put because the client is exposed to risks specific to the stock as well as overall market movements, whereas an index put (i.e. Nifty put) would only hedge broader market declines.

Assuming a premium of approximately ₹100 per share, the total premium cost for 500 shares would be around ₹50,000. The main risks of this strategy include theta decay, where the option loses value over time; wrong strike price selection, which may reduce hedge effectiveness; and liquidity risk if trading volumes are low.

Reflection :-

This week's contents helped me get a very good intuition into the exact terminology of options and what the option prices of some large cap stocks might be and how good they are in predicting market trends. I feel like the concepts covered till now have been engraved enough into my brain for me to not have to visit these videos again. The [video](#) on what exactly constitutes the time value of an option contract was really insightful and I feel like I would be keen to read more about it in the upcoming days.

References –

1. [Varsity by Zerodha](#)
2. [ProjectOption YouTube Channel](#)